insulin

Warnings	Classification	Alternate Names;
high alert medication (see Potential Hazards section)	hypoglycemic agent	different insulin types are NOT interchangeable

Indications

- treatment of diabetes mellitus
- diabetic ketoacidosis or coma
- intensive control of glucose in critical care patients
- treatment of severe hyperkalemia
- high-dose insulin antidote for hemodynamic support in some overdoses (e.g. calcium channel or beta blocker overdoses) contact **BC Drug and Poison Information Centre (DPIC): 604-682-5050 or 1-800-567-8911**

Insulin products (single ingredient) - onset and duration of action for subcutaneous route:

Insulin type	Generic name Insulin human (route)	Brand names of vials/pens	Onset of action	Peak effect	Duration of action	Administration timing
rapid acting	insulin lispro (subcutaneous)	HUMALOG ADMELOG	10 to 15 minutes	1 to 2 hours	3 to 5 hours	give just before meals
short acting	insulin regular (subcutaneous, IV)	HUMULIN-R NOVOLIN ge TORONTO	30 to 60 minutes	2 to 3 hours	6 to 8 hours	give 30 minutes before meals
intermediate acting	insulin NPH (subcutaneous only)	HUMULIN-N NOVOLIN ge NPH	1 to 3 hours	5 to 8 hours	up to 18 hours	as scheduled
long acting	insulin glargine (subcutaneous only)	LANTUS BASAGLAR	2 to 4 hours	no peak	20 to 24 hours	as scheduled

Reconstitution and Stability

- keep unopened vials, premix bags refrigerated (in original packaging to protect from light); do not freeze
- multidose vials and pens may be stored for up to 28 days at room temperature
- insulin regular (MYXREDLIN) premix bags: may be removed from the original packaging and stored at room temperature up to 25°C for up to 25 days. Once stored at room temperature, do not return to refrigerator.
- insulin suspensions - gently shake before use

Compatibility

- insulin regular human is compatible with sodium chloride 0.9%, dextrose, and lactated ringers

Preparation and Administration

Administration Route	Approved	Preparation and Administration Instructions	Required Monitoring
Subcutaneous	YES	rotate injection sites	Basic Monitoring
Intramuscular	NO		
IV direct	YES	use insulin syringe with Luer Lock tip insulin regular only; undiluted over 1 minute	Basic Monitoring
IV intermittent	NO		
Continuous IV infusion	YES	insulin regular only **standard solution concentration:** 1 unit/mL ▪ dilute 100 units in NS 100 mL ▪ or MYXREDLIN premix bag may be supplied ▪ *high-dose antidote indication*: dilute 100 units in NS 100 mL dilute 250 units in NS 250 mL *rate must be controlled with an IV pump*	Basic Monitoring *high-dose antidote* Full Monitoring See site-specific restrictions
intraperitoneal	YES	insulin regular only	Basic Monitoring

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insulin

Dosage

diabetes mellitus:
- subcutaneous dose is individualized according to blood glucose level
  - insulin regular: administer 30 minutes before each meal
  - insulin lispro: administer within 15 minutes of the meal
- continuous ambulatory peritoneal dialysis: insulin regular 5 to 15 units into each dialysis bag to maintain fasting blood glucose 5 to 8 mmol/L

diabetic ketoacidosis:(regimens may vary)
- insulin regular 0.15 unit/kg IV direct (optional) followed by insulin regular 0.1 unit/kg/hour continuous IV infusion titrated to optimize blood glucose level

intensive control of glucose in critical care:
- initiate insulin regular continuous IV infusion per treatment protocol

hyperkalemia:
- 25 to 50 g dextrose (50 to 100 mL dextrose 50%) IV direct over 5 minutes, followed by 5 to 10 units insulin regular IV direct or as per treatment regimen. Monitor blood glucose at 30 and 60 minutes post-dose and PRN for 6 hours.

high-dose insulin antidote for hemodynamic support in some overdoses:
- 1 unit/kg IV direct, followed by 0.5 to 1 unit/kg/hour continuous IV infusion. May take up to 60 to 90 minutes after initiation of therapy to see response. Doses as high as 10 units/kg/hour have been used.
- **must** be given with **dextrose**:
  - start dextrose infusion 0.5 g/kg/hour (e.g. 5 mL/kg/hour of D10W)
  - if initial blood glucose is less than 14 mmol/L give 25 g (50 mL of dextrose 50%) IV direct [see dextrose 50% monograph]
  - dextrose supplementation may be required for up to 24 hours after high-dose insulin is stopped.

Potential Hazards of Parenteral Administration

- lipodystrophy may occur at injection sites in up to 3% of users; rotate injection sites routinely and do not administer via subcutaneous butterfly
- high alert medication:
  - if given by IV route
  - for insulin injections at concentrations above 100 units/mL

Important Implications

Contraindications/Cautions
- *caution*: local allergic reactions (such as itching, redness, swelling, stinging or warmth at the injection site) may occur within the first 1 to 3 weeks and usually disappears within a few weeks; true insulin allergy is rare
- Pregnancy/Lactation: refer to Lexicomp or Micromedex

Side effects: hypoglycemia

Monitoring:
- blood glucose, signs of hypoglycemia, potassium
- *high-dose antidote*
  - blood glucose Q15 to 30 MIN until stable, then Q1H until several hours after insulin discontinued
  - potassium Q1H for at least 4 hours, then Q2H until insulin discontinued; magnesium and phosphorus

Other
- insulin regular or lispro can be mixed with an intermediate acting insulin from the same manufacturer; draw up insulin regular or lispro first. Inject mixture immediately
- insulin glargine must never be mixed with any other insulins
- Fixed ratio combinations of rapid or short acting insulin with an intermediate acting insulin also available

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